Project Ideas

Repo: https://github.com/jgre138/CSSE-2026-27-senior-project
Arduino/ESP32 Simulator/Builder https://wokwi.com/

https://wokwi.com/projects/475791115642232833 ESP32 analog circuit
How Ultrasonic Sensor Works: https://www.youtube.com/watch?v=iwD9RiqIHQw

Tinkercad: https://www.tinkercad.com/

https://www.reddit.com/r/AskABlindPerson/

Note: Allowed to use AI but make sure that you show that you had input and understand those changes

Remember: This doesn't need to be an issue that the world or everyone experiences; this can be something small that a few people have issues with. Just like my summer project, can be something for small quality of life changes.

- Glasses to help blind people (Help them avoid dangerous situations)
    - Potential Name: Third Eye, Miracle Glasses
    - Like the meta glasses, but instead of having to prompt it for things, it will keep checking surroundings. (this is important for business value!!)
    - Glasses with a camera, and 4 points of contact that will vibrate to be able to inform of what direction
    - possibly using earbuds (Don't want to obstruct their hearing, one of their important senses)
    - Recording to analyze and then delete data once it's not needed
    - How will the device be powered or charged?
    - Look at current technology for our target audience and stuff with similar technology of what we want to make.
        - Not sure how relevant this will be, but here's a link to hardware specs of an ultrasonic walking stick for the blind that may give us some ideas for the hardware of the glasses: https://nevonprojects.com/ultrasonic-blind-walking-stick-project/

- Something to highlight items in line of sight. ?
- A program where you input a song and it outputs sheet music/chords/etc for a selected instrument

    - Example: you put in a song, and then the program will identify what instruments are playing. From those, you can select which of those instruments you want sheet music/chords for and the program will generate it
- ---
  Hey guys this is Yu Kun Cai

Based on my understanding, our project is an **AI-powered smart glass to help visually impaired individuals navigate safely**. To make it more structured for our upcoming proposal, I think we can break it down into four core modules:

1. **Obstacle Detection:** Combining ultrasonic sensors for distance and a camera with light AI models for proactive object identification.
2. **Feedback System:** Audio alerts via open-ear/bone-conduction headphones (so we don't block their hearing) and 4-point haptic vibration on the frames for directions.
3. **Power Management:** Exploring portable, rechargeable, and low-power consumption solutions tailored for wearables.
4. **Safety & Privacy:** Ensuring a secure physical fit (anti-slip design) and implementing a 'delete-after-analysis' data privacy protocol.

Let me know if this aligns with everyone's thoughts! I can take the lead on researching one of these modules if you like. Am I right? By the way, I have a soldering iron and some basic soldering experience, so I can definitely help with the soldering part when we start putting the electronic components together.

Once we all agree on the hardware components and get them ready, we can work on the physical assembly together as a team!

Brain and Video Camera

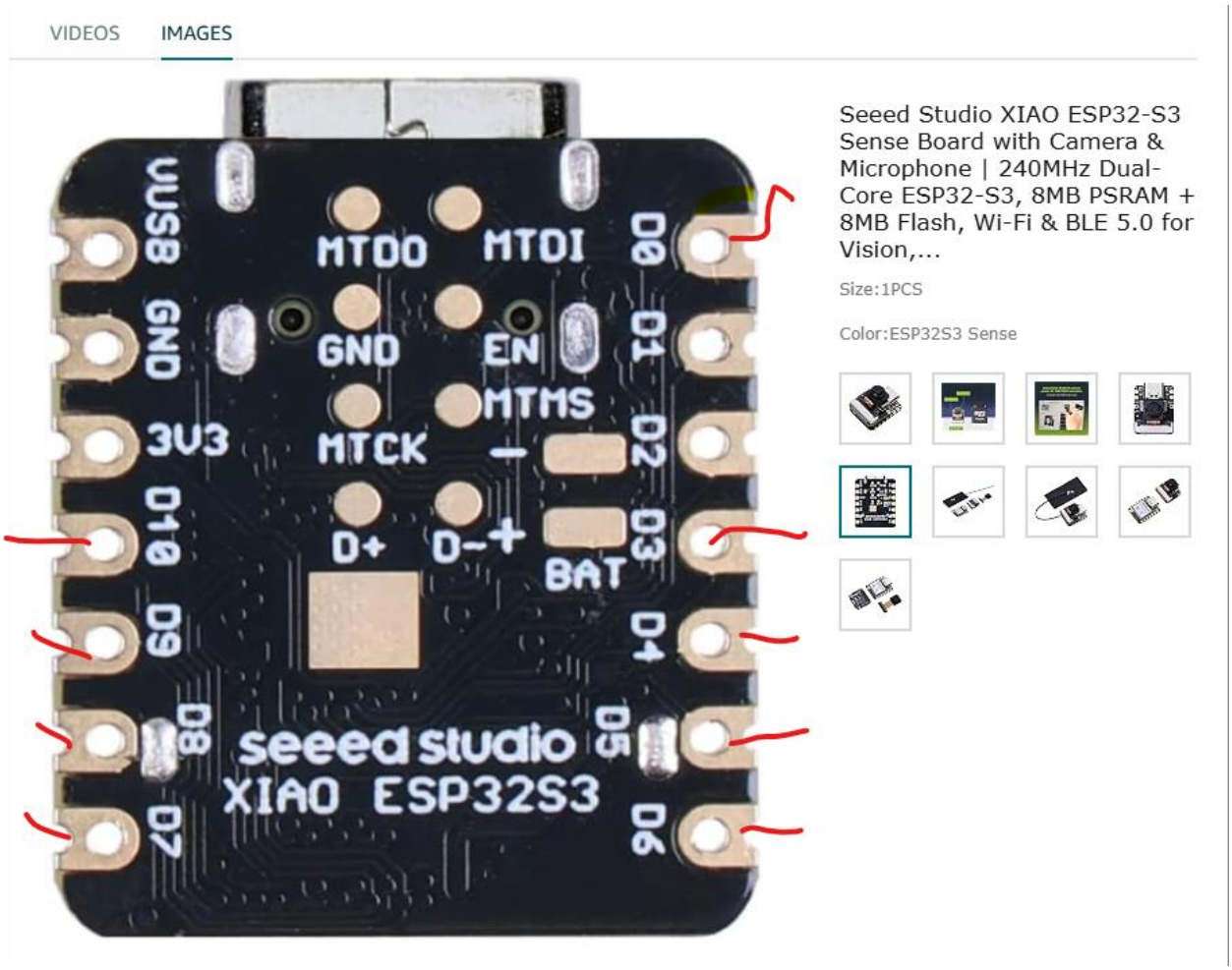
https://www.amazon.com/gp/aw/d/B0C69FFVHH?pd_rd_plhdr=t&hsa_cr_id=0&qid=1789396775&sr=1-1-fec03104-6f11-4242-8f79-15fae2896f66&i=aps&aref=MJgsG1fpNU&_encoding=UTF8&ref_=sbx_s_sparkle_sbtcd_asin_0_img&pd_rd_w=12Dbc&content-id=amzn1.sym.22e060df-ce87-467b-805a-61b8bd37fbe9%3Aamzn1.sym.22e060df-ce87-467b-805a-61b8bd37fbe9&pf_rd_p=22e060df-ce87-467b-805a-



This motherboard already includes a camera.

Vibrator, motor

61b8bd37fbe9&pf_rd_r=JWPTAGHVTMCGP2QJF4ZM&pd_rd_wg=yZH8C&pd_rd_r=d8aca39b-558a-48f8-b37a-af6fd87cfb46&th=1



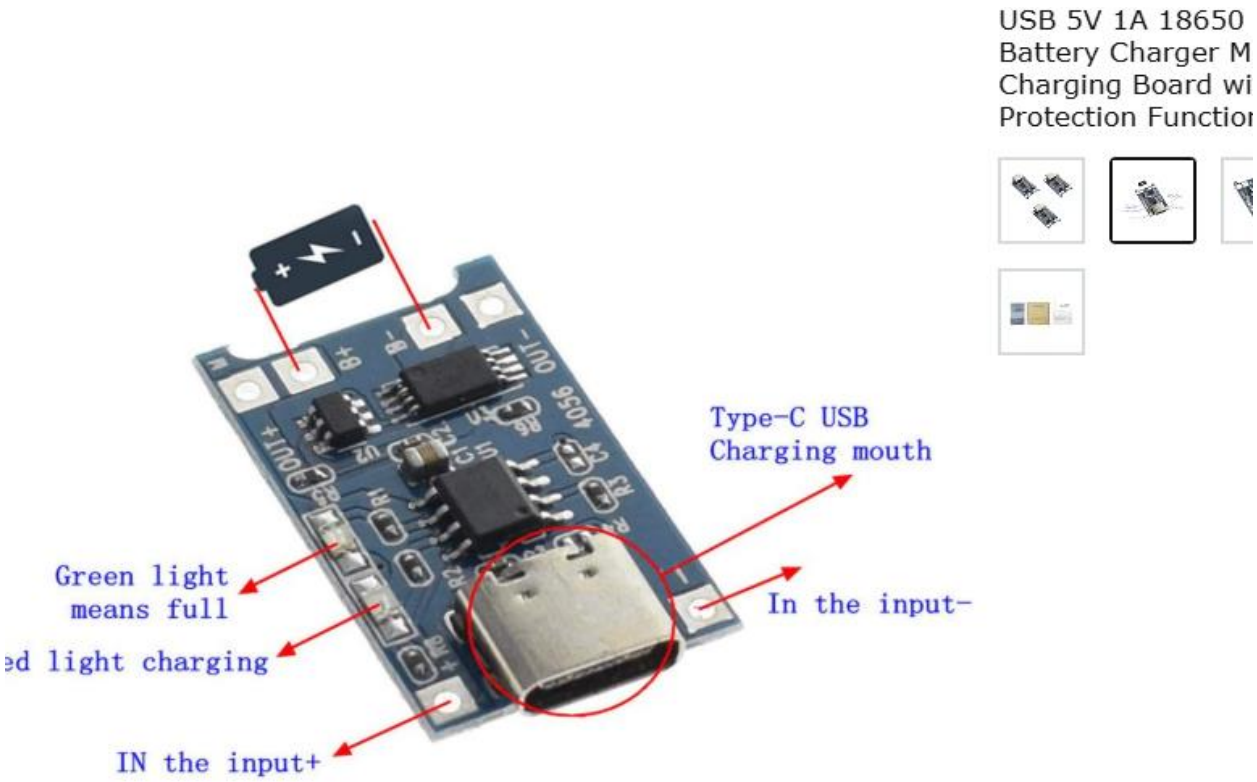
We can make 9 motor on this board



This is a vibration motor.

Charging circuit board

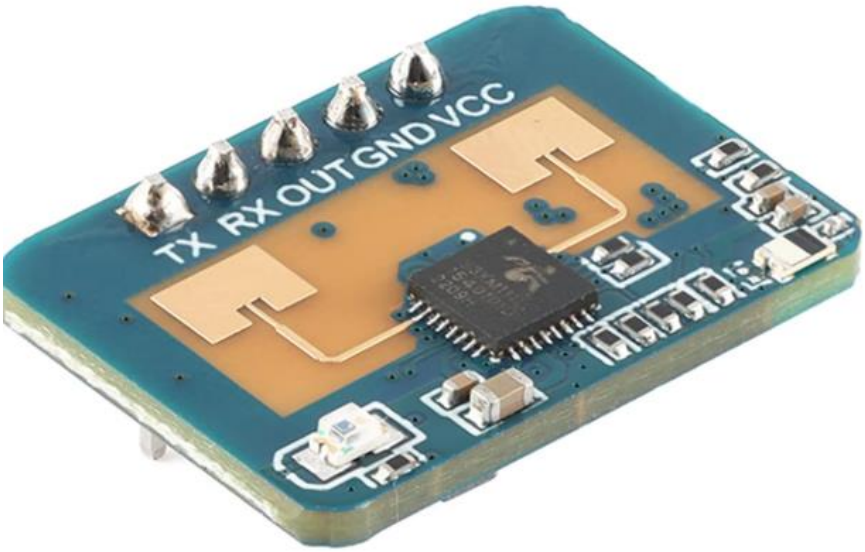
https://www.amazon.com/HiLetgo-Lithium-Charging-Protection-Functions/dp/B07PKND8KG/ref=sr_1_1_sspa?dib=eyJ2IjoiMSJ9.-UiBtZJmSZvxE3u_ela8FVY6JBEU5R6BcJjnIbdCHaDCSCMsGmFG2cV5LC0SKkff_odvcF-wyTb2mCzKjkSSoMmE7f1zVk01fPJxzA5RjnKiOyaRUuiqYNBW5qKSxgBfy17iRDh_qqUx3DujhSWgYIBL4Wch3jyDdDHslaptwIdok2eFUw_5qSpaizmbuDPZLa6AQvZ09DidJ5zGVmASesLq64ruPEW0VWF8m58HkXg._Y9Fht4L3jZKqHtrIuSpFd63GjaT6yUez-Pf0Rco2JI&dib_tag=se&keywords=TP4056&qid=1789396790&sr=8-1-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&psc=1



This is a charging template. Depending on the situation, you can use a charging battery or directly connect an ordinary battery without charging.

Radar detection

https://www.amazon.com/dp/B0BXDLHHH2?ref=clp_hp_h_pc





This is the radar detection panel. According to official information, it can detect up to 5 to 6 meters at the farthest distance.

Sensor VS Camera Research

https://www.hokuyo-usa.com/lidar-vs-cameras-vs-ultrasonic-sensors

- It seems to me that we might not need cameras at all if this is true. LiDAR + Sensor might be the move
- Might not even need LiDAR – ultrasonic sensors are good for short-range object detection while LiDAR is better for long-range. We don't really need to worry about long-range as much if we're just trying to avoid collision with nearby objects

Options for sensors:

- Ultrasonic Sensor
    - https://www.sparkfun.com/catalog/product/view/id/8147/?srsltid=AfmBOorsuLd76TZYi5H2MYjkz6Z9IcxDp39ugjmCRXHUF0qY4_5bUOgKs8E (Here is an option that has a range of around 13 feet, not as far as lidar or camera, but much more cost efficient)
- ir infrared obstacle avoidance sensor
- Ultrasonic Parking Sensor
- LiDAR

Esp32 board as the brain.

- Can program in C++ in Arduino IDE or VSCode

New Question: How do we get sensor data from the brain to our computers when the device is not connected directly?

Two options: Wifi or Bluetooth

- One concern is if using wifi, cellular data cannot be used as the phone is connected to a wifi network despite there being no internet.
    - Bluetooth would avoid that issue, just need to see how to have the esp32 use bluetooth and how much data it can send at once
    - This would be for testing, unsure if for production.
    - https://randomnerdtutorials.com/esp32-bluetooth-guide/

Another Question: Is an app needed at all?

Feedback System:

- Buzzer?? (https://www.amazon.com/dp/B07XDPXH7K?lv=shuf&channelId=500&plpRedirect=mhFallback) as an example

    - The device itself can beep and vibrate when an obstacle is near and then also send a message to the headphones ?
- Directional feedback : maybe a beep on either the left or right side of the headphones?

Current Ideas:

- Glasses and belt clip
    - Belt clip holds the brain and/or sensors
- Bone Conducting earbuds and Belt Clip

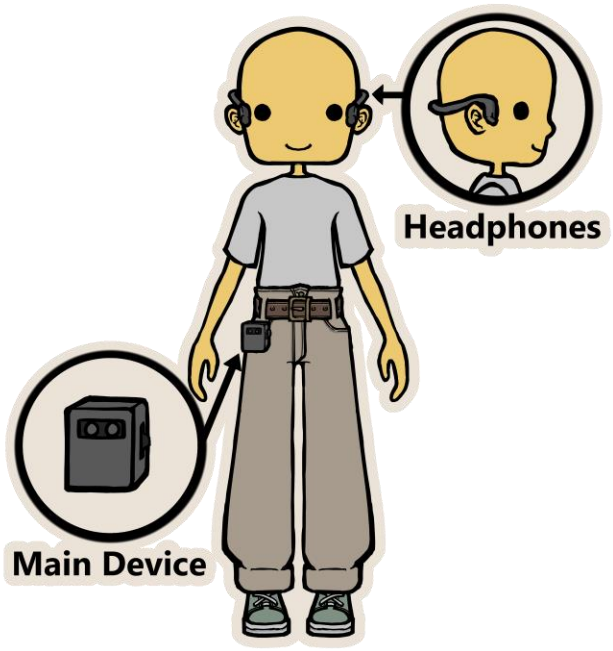
Glasses route:

- The glasses hold vibration modules or buzzers.
    - Will need a battery to power the sensors and some module to be able to send the data to the esp32. And have the esp32 send requests to have the buzzers go off.
    - May hold the sensors or a camera is we decide to add one
- The belt clip will hold a battery, the brain (esp32) , other parts
    - This could hold the sensors as well

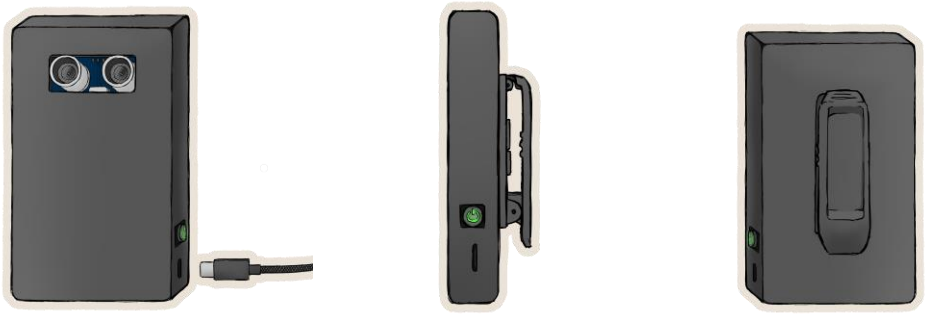
Headphones route:

- *All* hardware would be contained in the main device clipped to the belt and have access to power from the battery in there
- Headphones connect via bluetooth to provide audio feedback/ specific directional cues if connected. Headphones are not required, just an option to provide additional feedback.

- Diagram 1 – User model



- Diagram 2 – Main device outer shell



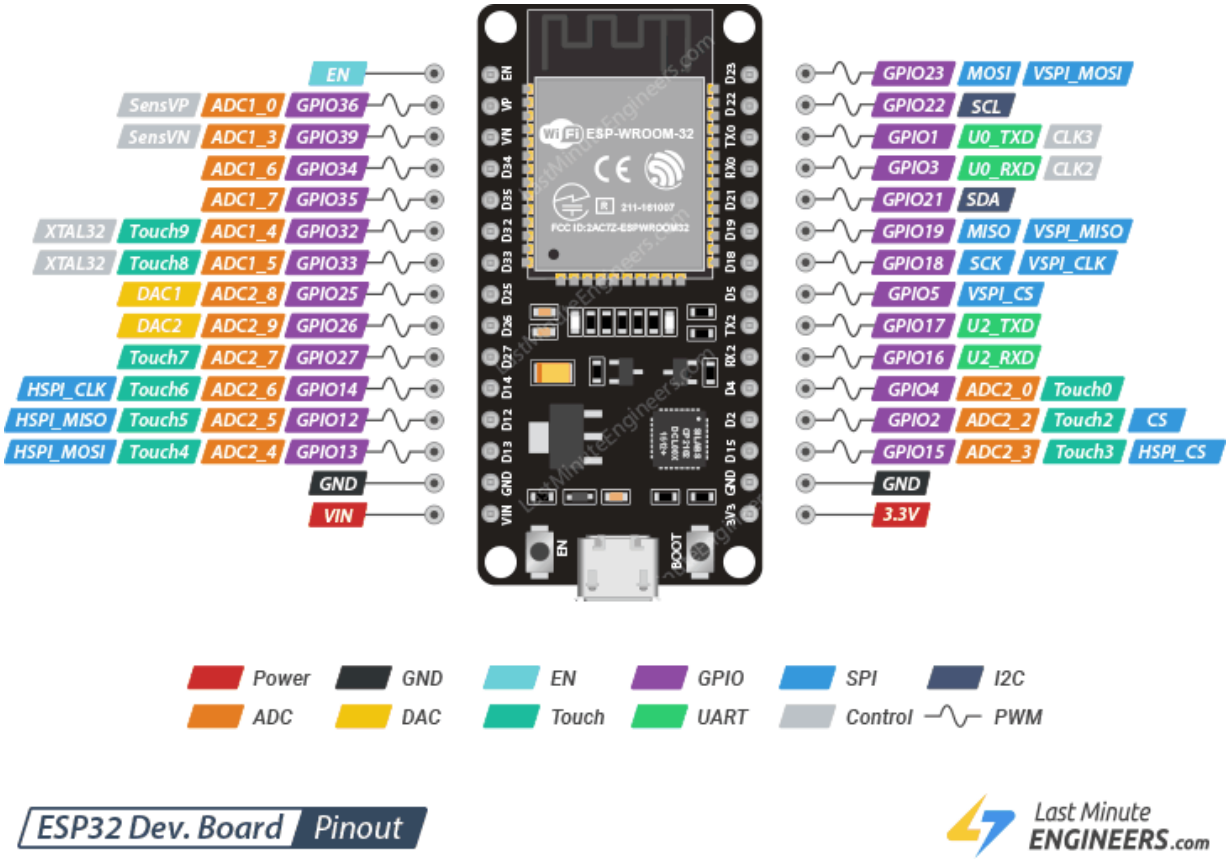
Front Side Back

- EVERYTHING is in the main device – main processor, buzzer/vibration motor, ultrasonic sensor
    - When an obstacle is detected in the immediate vicinity, the device will vibrate (maybe also make a loud beeping sound) to alert the user
    - Bluetooth headphones are a separate device that may or may not be connected for specific audio cues. Not necessarily required
        - If headphones are connected via bluetooth, then additional directional cues or messages may be provided

Overall: A belt clip will be used to hold the brain of the product, and the battery??

Write the code in either C++ or Python

Battery ideas:

- Rechargeable via usb C by adding a charging module (like a TP-4056) powered by a lithium ion battery? **NOTE:** we will probably have to worry about voltage if we go this route. I have a multimeter to read voltage if needed
    - ESP32s use a voltage of 3.0v to 3.6v, and you do NOT want to go above that.
    - Lithium Coin batteries may get this voltage. Or LiFePO4 which is 3.3v and safe.
    - We CAN use a 3.7 Lithium ion battery as long as its connected to a development board
    - 

    - ESP32 Dev. Board Pinout — Last Minute ENGINEERS.com

Sept 21:

Three ideas to investigate:

1. Glasses and Brain box

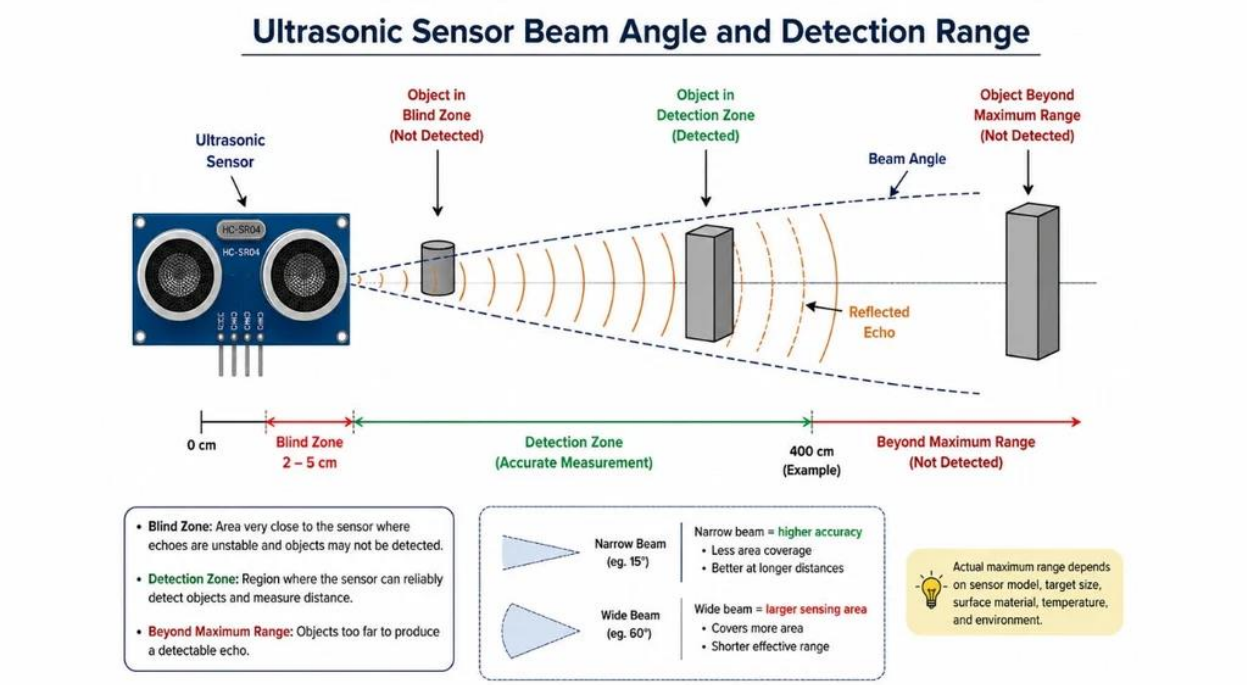
    a. Glasses with the sensors, vibration motors, and sound (?), battery for power
    b. Brain box with the microcontroller and battery
2. Bone conducting earbuds and brain + sensor box
    a. Earbuds for optional audio feedback
    b. Brain box has the sensors and the battery.
3. Microcontroller (ESP32) Vs. Raspberry Pi

This semester we focus on a proof of concept and demo of a working product. Not a final product that is ready for production. Pretty much we need a prototype.

I was looking at meta glasses online to see what they use for image recognition and I found this: https://www.meta.com/ai-glasses/blind-visually-impaired/?srsltid=AU7gw4UpDf9V4vyrrVt1JHtewGbyXRZ63GoX9iKJEkYas-jEz337m0K5

(This seems similar to what we were planning, we should try our best to distinguish what we are working on to this already existing technology)

- These seem specifically more geared towards object identification and text-to-speech, which if I'm right is not exactly what we're focusing on
- We're trying to make a device that uses sensors to prevent the user from tripping over/walking into obstacles around them. In this case, we do not need cameras or object detection that specific – IE we don't need to know what the object is, just that it is there and it is a hazard that the user can trip over or bump into.
    - Ultrasonic sensors should be sufficient for this. They use sound waves to be able to detect objects and even distance. (Example: like how back-up cameras in cars work – you get too close to an object, it'll get set off and alert you so you can brake and not hit anything)



1. Arduino Uno: https://amzn.to/4f9FjeZ 2. Buzzer: https://amzn.to/48nfczd 3. Ultrasonic sensor: https://amzn.to/40d60LD 4. Vibration motor: https://amzn.to/4hcsfr0 5. Jumper wires: https://amzn.to/489aRiO

https://www.youtube.com/watch?v=qtlgMRC9qCo

https://www.amazon.com/2-Pack-Battery-Module-Rechargeable-1200mAh/dp/B0GWZQFK8T?dib=eyJ2IjoiMSJ9.77Kk2Qcq54xO9_ovLaPeZ1G_iWWqOPwBF_iWPaJnQCFA4RtKubzkGtWXXosMJe-1Zk0QQs_il-2_Y5WNC0LQbuQ6xU5O8KvUvD2Ck7QhoiYUMpqz9R4DHzl2GCYKPRPBlKowL0IiYXCbjfF7bvHIdcovhl6ZJkJBCFz20YnwazTJTVvP6lQx1NVvjdWJcTfMVu1s6sW4xO9nuJpIpF-FWVu8lv60f_Y06MHQuFFUxGieuegfSxY0oz1ybUZZm3Tb9KvYi3ZqOaECvswe2XqZg_I6iuI3S3Ua3tJdrz-vifk.CE8v-yfI-OqTO9-6CLSbU7qDzLssDpet3up7ABa1ir8&dib_tag=se&keywords=5v%2Brechargeable%2Bbattery%2Bpack&qid=1790175046&sr=8-15&th=1

https://framos.com/articles/what-are-depth-sensing-cameras-and-how-do-they-work/ (Interesting article about some other image processing options we could explore if the sensors dont work out)

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1. Core Controller & Sensors

- **ESP32 Development Board (2-Pack)**
  - 🔍 **Amazon Search Key:** `ESP32 development board 30pin pre-soldered`
  - ⚠️ **Note:** We must get the **pre-soldered** version so it can plug directly into our breadboards without manual soldering.
- **Ultrasonic Distance Sensors (5-Pack)**
  - 🔍 **Amazon Search Key:** `HC-SR04 ultrasonic sensor 5 pack`
  - ⚠️ **Note:** This standard 4-pin blue version perfectly fits the 5 directions of our waist-mounted radar array.

2. Headband Haptic Feedback Components

- **Coin Vibration Motors (10-Pack)**
  - 🔍 **Amazon Search Key:** `3V 5V coin vibration motor 10mm`
  - ⚠️ **Note:** These flat mini motors usually come with adhesive backing, which is perfect for mounting inside the 3D-printed headband pockets.
- **Active Buzzer (10-Pack)**
  - 🔍 **Amazon Search Key:** `5V active buzzer piezo`
  - ⚠️ **Note:** Make sure it is labeled **Active**. Active buzzers will beep instantly when powered, which matches our non-blocking variable frequency alert code.

3. Circuit Components for Motor Drivers (Scheme B)

- **Electronic Component Assortment Kit (Transistors & Diodes)**
  - 🔍 **Amazon Search Key:** `Electronic component assortment kit TO-92 transistor diode`
  - 📋 **Note:** Buying an assortment kit is way cheaper than buying bulks separately. Please double-check the product description to ensure it includes **2N2222** (or BC337) NPN transistors and **1N4007** diodes.

4. Wearable Cable & Power Supply

- **Flexible Long Cable for Headband (7-Wire Connection, > 1 Meter)**
  - 🔍 **Amazon Search Key:** `26 AWG ribbon cable 10 colors 5m` OR `Cat6 ethernet cable 10ft`
  - ⚠️ **Note:** The wire MUST be **Stranded** (flexible silicone or patch cable) rather than solid core, so it won't snap when the user moves or turns their

head. If we buy a Cat6 cable, we can just strip the outer jacket to get 8 color-coded flexible wires inside.

- **Waist-Box Power Supply**
  - 🔍 **Amazon Search Key:** `4 AA battery holder with DC jack and switch`
  - 📋 **Note:** 4 AA batteries provide around 6V, which can stably power the ESP32 via the VIN pin and easily drive 5 vibration motors simultaneously.

Hey guys this is the parts wo might gonna need for our project, I was ask GPT to give me which parts I need to get , I haven't checked these parts online yet. I would like to explain that some of the transistors, diodes, resistors and other components needed here are added to protect the circuit. Otherwise, the ESP32 might be damaged by the sudden increase in current.
